

Yelp Review Intelligence

End-to-End NLP Pipeline for Restaurant Review Intelligence

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1.Executive Summary

Customer reviews contain valuable insights into customer satisfaction, operational issues, and service quality. However, manually analyzing millions of reviews is neither practical nor scalable for businesses operating across multiple locations.

This project presents an end-to-end Natural Language Processing (NLP) solution that transforms unstructured Yelp reviews into actionable business intelligence. Rather than performing sentiment classification alone, the proposed system enriches each review with complaint topics, severity assessment, business recommendations, and dashboard-ready outputs.

The solution was developed using the Yelp Academic Dataset, containing nearly **7 million reviews**, enriched with business, user, and tip information. To evaluate different modeling strategies, three approaches were implemented and compared: **TF-IDF + Logistic Regression**, **DistilBERT**, and **TextCNN**. The final model was selected by considering not only predictive performance but also inference speed, deployment complexity, scalability, and operational cost.

The resulting system includes a complete machine learning pipeline covering data preparation, feature engineering, model training, REST API deployment, and business intelligence generation, providing a practical foundation for production-ready review analytics.

2.Business Problem

Online review platforms generate an enormous amount of customer feedback every day. While these reviews contain valuable information about customer satisfaction and operational performance, their volume makes manual analysis impossible for restaurant managers and business owners.

Traditional sentiment analysis only identifies whether a review is positive or negative, providing limited business value. Managers also need to understand why customers are satisfied or dissatisfied and which operational areas require attention.

For this project, the primary stakeholder was defined as a Restaurant Operations Manager responsible for monitoring customer experience across multiple restaurant locations. The

objective was to develop an AI-powered review intelligence system capable of automatically transforming free-text reviews into structured business insights.

Instead of returning only a sentiment prediction, the proposed solution generates:

- Sentiment classification
- Complaint topic detection
- Severity assessment
- Confidence score
- Recommended business actions

This enables decision-makers to quickly identify recurring issues, prioritize operational improvements, and monitor customer satisfaction without manually reviewing thousands of customer comments.

3.Dataset

This project is based on the Yelp Academic Dataset, a publicly available dataset containing millions of customer reviews together with rich contextual information about businesses, users, and customer interactions.

Unlike many sentiment analysis datasets that only contain text and labels, Yelp provides multiple related data sources, allowing the integration of contextual information into the modeling process.

The project uses the following raw datasets:

Dataset	Description	Records
Review	Customer reviews including review text, rating, date and engagement statistics	6,990,280
Business	Business metadata including categories, location and ratings	150,346
User	Reviewer information and historical activity	1,987,929
Tip	Short customer tips posted independently from reviews	908,915

The four datasets were merged using the common `business_id` and `user_id` keys to create a unified analytical dataset.

The resulting modeling dataset contains approximately 7 million customer reviews enriched with contextual business and user information.

a. Available Features

The original Yelp datasets provide a diverse collection of structured and unstructured variables.

Review-level information

- Review text
- Star rating (1–5)
- Review date
- Useful votes
- Funny votes
- Cool votes

Business information

- Business name
- Categories
- City
- State
- Geographic coordinates
- Overall business rating
- Number of business reviews
- Opening status
- Business attributes

User information

- Number of historical reviews
- Average user rating
- Yelp membership duration
- Elite membership
- Number of fans
- Historical useful/funny/cool votes

Tip information

- Tip text
- Tip date
- Number of compliments

b. Data Integration

The raw JSON files were processed using **DuckDB** together with **Pandas**, allowing efficient handling of nearly seven million reviews without requiring a distributed computing framework.

To prevent information leakage, historical information was carefully aggregated using only information available before the review date.

For example:

- Tips were aggregated only from records created before each review.
- Business statistics were treated as historical metadata.
- User experience features were computed using historical user information.

This ensured that no future information was introduced during model training.

c. Final Modeling Dataset

After preprocessing and feature engineering, each review consists of:

- Raw review text
- Historical tip information
- Business category
- Business location (city/state)
- Business statistics
- User experience statistics
- Review length features
- Temporal information
- Target sentiment label

The final dataset contains approximately **6.99 million observations** with **32 engineered features**, making it substantially richer than a conventional text classification dataset.

4. Exploratory Data Analysis (EDA)

4.1 Purpose of the EDA

Before building any machine learning model, an extensive exploratory data analysis (EDA) was conducted to better understand the Yelp dataset and identify patterns that could improve sentiment prediction.

Due to memory limitations, the complete dataset of nearly **7 million reviews** could not be loaded into memory simultaneously. Therefore, the EDA was performed on a representative

subset consisting of **3 million randomly selected reviews**, which still provides statistically meaningful insights because of its large size.

The objectives of the EDA were:

- Understand the quality of the merged dataset
- Analyze review text characteristics
- Examine class distribution
- Investigate relationships between review sentiment and contextual features
- Identify potential predictive signals for feature engineering
- Detect possible data quality issues before modeling

4.2 Missing Value Analysis

The first step was to examine missing values after merging the Review, Business, User and Tip datasets.

Only a very small percentage of missing values was observed.

The highest missing ratio (approximately **7%**) belongs to tip-related features, which is expected because not every business receives customer tips.

Business attributes have approximately **3%** missing values, while the remaining variables are almost complete.

This indicates that the overall data quality is very high and no aggressive missing value treatment was required.

column	missing_ratio
latest_tip_date	0.07
avg_tip_compliment	0.07
sample_tips	0.07
tip_count	0.07
attributes	0.03
categories	0.00
elite	0.00
fans	0.00
cool	0.00
funny	0.00

Figure 4.1 — Missing Value Ratios

4.3 Review Length Analysis

Since text length often affects NLP model performance, review lengths were analyzed in detail.

Summary statistics show:

- Average review length: **104.7 words**
- Median review length: **75 words**
- Maximum review length: **1,059 words**
- Maximum character count: **5,000 characters**

This indicates that Yelp reviews vary significantly in size, ranging from extremely short comments to very detailed customer experiences.

	review_word_count	review_char_count
count	3,000,000.00	3,000,000.00
mean	104.66	567.17
std	97.70	526.25
min	1.00	1.00
25%	42.00	229.00
50%	75.00	405.00
75%	133.00	720.00
max	1,059.00	5,000.00

Figure 4.2 — Review Length Statistics

The histogram below confirms that the distribution is highly right-skewed. Most customer reviews contain between **20 and 150 words**, while only a relatively small proportion are very long reviews. To keep training computationally efficient while preserving nearly all semantic information, review texts were truncated to **300 words** during preprocessing.

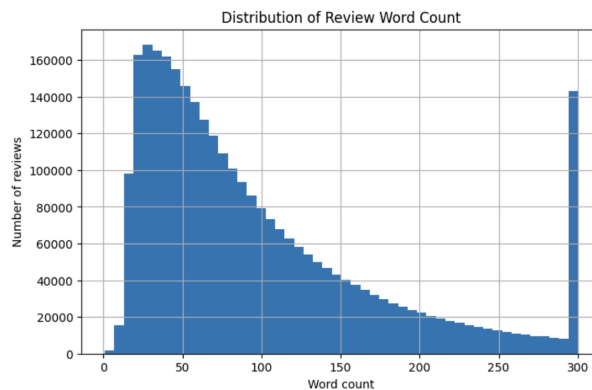


Figure 4.3 — Distribution of Review Length

4.4 Rating Distribution

The target variable consists of the original Yelp star ratings. The distribution is clearly imbalanced toward positive reviews.

Approximately:

- **46%** are 5-star reviews
- **21%** are 4-star reviews
- **15%** are 1-star reviews
- **10%** are 3-star reviews
- **8%** are 2-star reviews

This reflects the natural tendency of online review platforms where satisfied customers submit reviews more frequently.

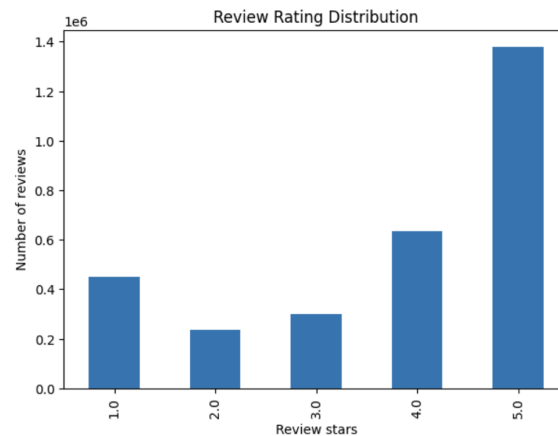


Figure 4.4 — Rating Distribution

Although the classes are imbalanced, they still contain hundreds of thousands of observations each, making them suitable for supervised learning without requiring oversampling techniques.

4.5 Review Length vs Rating

One interesting observation is the relationship between review length and customer satisfaction.

Negative reviews tend to be considerably longer than positive ones.

Average review length by rating:

Rating	Average Words
1	133.6
2	133.8
3	123.9
4	107.8
5	84.6

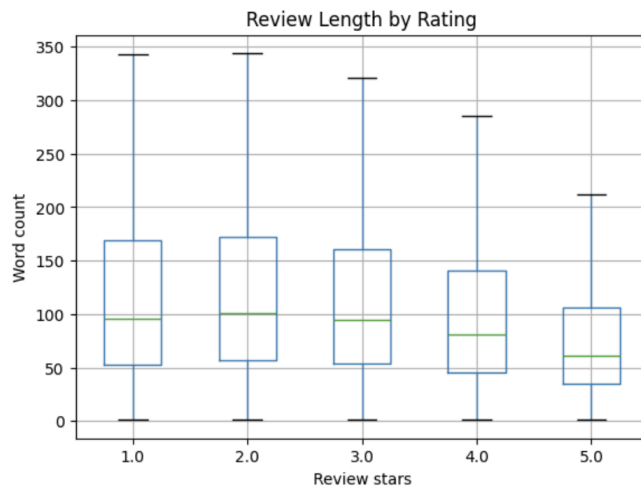


Figure 4.5 — Review Length by Rating

This finding suggests that dissatisfied customers usually provide detailed explanations about problems they experienced, whereas satisfied customers often leave shorter comments such as "Great food" or "Excellent service."

As a result, **review length** was included as an engineered feature in the final models.

4.6 Most Informative Words

A TF-IDF analysis was performed to inspect the most influential words associated with positive and negative reviews.

The most important negative terms include:

- service
- order
- told
- time

- food

while positive reviews are dominated by words such as:

- great
- amazing
- delicious
- best
- love

term	score
food	0.03
service	0.03
just	0.02
place	0.02
time	0.02
like	0.02
order	0.02
told	0.02
don	0.02
said	0.02

term	score
great	0.04
food	0.03
good	0.03
place	0.03
service	0.02
best	0.02
time	0.02
love	0.02
delicious	0.02
amazing	0.02

Figure 4.6 — Most Important Negative Terms | Figure 4.7 — Most Important Positive Terms

These findings indicate that customer satisfaction is primarily driven by **service quality**, **food quality**, and **overall experience**, validating the use of textual information as the primary modeling feature.

4.7 Category-Level Analysis

Restaurant categories exhibit noticeably different customer satisfaction profiles. The largest categories, including Restaurants, Food, Bars and Nightlife, generally receive ratings close to four stars. However, several service-oriented categories show consistently poor customer satisfaction.

For example:

Category	Average Rating	Negative Ratio
Television Service Providers	1.68	82%
Internet Service Providers	1.81	79%
Truck Rental	2.20	68%
University Housing	2.28	65%

category	review_count	avg_rating	avg_review_length	negative_ratio
Restaurants	2036484	3.79	100.39	0.21
Food	789607	3.90	97.34	0.18
Nightlife	672769	3.79	105.90	0.20
Bars	639794	3.79	105.44	0.20
American (Traditional)	435353	3.68	101.86	0.23
American (New)	408599	3.82	108.99	0.19
Breakfast & Brunch	364519	3.87	98.81	0.18
Sandwiches	298113	3.87	92.75	0.19
Event Planning & Services	263826	3.71	116.63	0.23
Seafood	256429	3.81	101.58	0.20

Figure 4.8 — Highest Rated Categories

category	review_count	avg_rating	avg_review_length	negative_ratio
Television Service Providers	1962	1.68	132.45	0.82
Internet Service Providers	3639	1.81	132.08	0.79
Truck Rental	1952	2.20	139.15	0.68
University Housing	1357	2.28	178.11	0.65
Mortgage Lenders	1026	2.42	127.22	0.64
Telecommunications	2484	2.50	126.39	0.61
Post Offices	2155	2.44	109.32	0.59
Property Management	5377	2.59	155.52	0.59
Banks & Credit Unions	3340	2.60	111.74	0.57
Insurance	2998	2.68	126.55	0.57

Figure 4.9 — Lowest Rated Categories

This analysis demonstrates that the business category provides meaningful contextual information beyond review text alone.

Therefore, business categories were incorporated into the feature engineering pipeline.

4.8 Geographic Analysis

Customer satisfaction also varies across cities.

Major metropolitan areas such as:

- New Orleans
- Santa Barbara
- Nashville

show relatively high average ratings. Conversely, several smaller cities consistently exhibit higher negative review ratios.

city	state	review_count	business_count	avg_rating	negative_ratio	avg_review_length
Philadelphia	PA	434583	7040	3.78	0.20	113.27
New Orleans	LA	274179	2975	3.96	0.16	100.59
Tampa	FL	191271	4325	3.74	0.24	104.19
Nashville	TN	185672	3301	3.81	0.21	104.27
Tucson	AZ	175600	4379	3.72	0.25	103.21
Indianapolis	IN	163363	3594	3.84	0.20	109.14
Reno	NV	150166	2835	3.75	0.24	110.50
Saint Louis	MO	111670	2398	3.86	0.19	110.78
Santa Barbara	CA	106250	1821	3.98	0.18	98.84
Edmonton	AB	45480	2463	3.61	0.22	134.39

Figure 4.10 — Highest Rated Cities

city	state	review_count	business_count	avg_rating	negative_ratio	avg_review_length
Fenton	MO	3129	136	2.80	0.50	118.26
Flourtown	PA	1087	28	2.85	0.48	112.23
Bridgeton	MO	1441	75	3.03	0.44	112.63
Fort Washington	PA	1219	32	3.12	0.41	128.24
Manchester	MO	1012	62	3.22	0.40	117.75
Bristol	PA	1192	60	3.15	0.40	119.86
Montgomeryville	PA	1101	49	3.17	0.40	113.50
Turnersville	NJ	1445	71	3.23	0.40	115.05
King Of Prussia	PA	3276	62	3.09	0.39	124.74
Warrington	PA	2778	95	3.19	0.39	103.92

Figure 4.11 — Lowest Rated Cities

These observations motivated the inclusion of **city** and **state** as contextual features in the final models.

4.9 User Behaviour Analysis

Reviewer experience was investigated using historical review counts. More experienced Yelp users generally provide slightly higher average ratings than first-time reviewers.

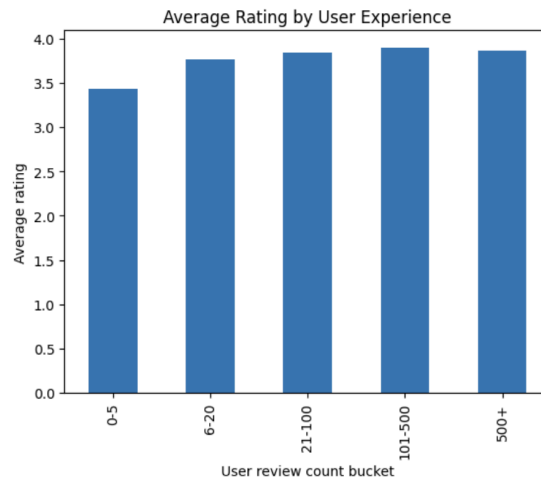


Figure 4.12 — Average Rating by User Experience

Although the effect is moderate, historical reviewer behavior contributes additional context for sentiment prediction.

4.10 Useful Reviews

Reviews marked as useful by other users are substantially longer than average reviews.

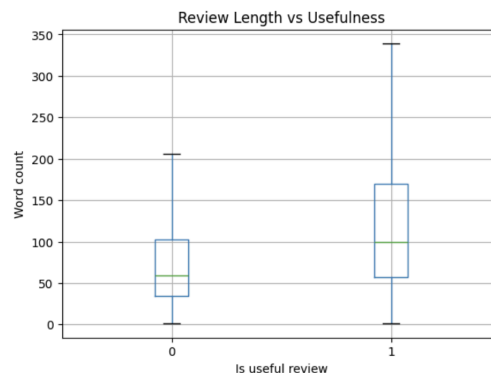


Figure 4.13 — Review Length vs Usefulness

This suggests that detailed reviews contain richer information and are considered more valuable by the community.

4.11 Temporal Analysis

Average customer ratings remain relatively stable over time. Small fluctuations are visible during the early years of Yelp due to the limited number of reviews, while the overall average stabilizes around **3.7–3.8 stars** as the platform matures.

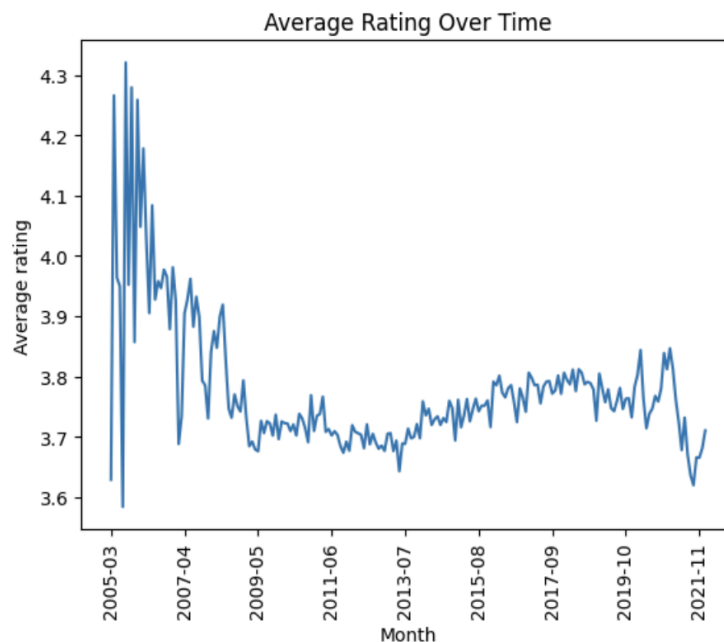


Figure 4.14 — Average Rating Over Time

Because review behavior evolves gradually over time, the final modeling pipeline uses a **chronological train/test split** instead of random sampling to avoid temporal data leakage.

4.12 Weekly Behaviour

Review patterns were also analyzed across weekdays. Average ratings remain highly stable throughout the week, ranging between **3.71 and 3.78 stars**, with only minor differences in negative review ratios.

review_weekday	review_count	avg_rating	negative_ratio
Monday	442748	3.73	0.23
Tuesday	405179	3.76	0.22
Wednesday	405427	3.77	0.22
Thursday	394011	3.78	0.22
Friday	403623	3.77	0.23
Saturday	456420	3.75	0.23
Sunday	492592	3.71	0.24

Figure 4.15 — Ratings by Weekday

This indicates that weekday effects are relatively weak and are unlikely to be strong predictors individually.

4.13 Effect of Customer Tips

Businesses that receive customer tips exhibit noticeably higher customer satisfaction.

Has Tips	Average Rating	Negative Ratio
No	3.59	32%
Yes	3.76	22%

Figure 4.16 — Businesses With vs Without Tips

This finding supports the hypothesis that customer engagement signals contain useful information beyond review text, motivating the inclusion of aggregated historical tip features in the final feature set.

4.14 Key Findings

The exploratory analysis provided several insights that directly influenced the subsequent feature engineering process.

The most important findings are summarized below:

- Customer reviews are generally long enough to contain rich semantic information.
- Negative reviews tend to be significantly longer than positive reviews.

- Business category strongly influences customer satisfaction.
- Geographic location contributes additional contextual information.
- Historical customer tips correlate with higher review scores.
- Experienced Yelp users exhibit slightly different rating behavior.
- Review text remains the strongest predictive signal, while structured contextual features provide complementary information.

These observations guided the design of the final feature engineering pipeline, which combines textual information with business, user, location and historical interaction features before training the sentiment classification models.

5. Feature Engineering

The raw Yelp dataset consists of four independent JSON files (Reviews, Businesses, Users, and Tips). These datasets were integrated into a single analytical dataset using `business_id` and `user_id` as the primary join keys.

Special attention was paid to preventing information leakage. Since the objective is to simulate a real production environment, only information available before the review date was used during feature generation. Historical customer tips were aggregated chronologically, ensuring that future information was never exposed to the model.

5.1 Data Integration

Each review was enriched with additional contextual information from the remaining datasets. The final record for every review contains:

- Review text
- Historical customer tips
- Business information
- User information
- Temporal information

The resulting dataset combines unstructured review text with structured business metadata, enabling the models to learn both linguistic patterns and contextual business characteristics.

5.2 Engineered Features

Several additional variables were created to improve predictive performance.

Text Features

- Review text
- Review word count
- Review character count

Business Features

- Business categories
- City

- State
- Average business rating
- Business review count

User Features

- User average rating
- User review count
- Yelp membership duration
- Elite status
- Historical useful/funny/cool counts

Historical Tip Features

- Only tips created before the review date were aggregated.

Generated features include

- Tip count
- Average tip compliments
- Sample historical tips
- Latest historical tip date

5.3 Feature Selection

Instead of assuming that all engineered variables improve performance, multiple feature configurations were evaluated through an ablation study.

The following feature sets were compared:

- Review text only
- Review + business categories
- Review + categories + historical tips
- Full contextual feature set

This experimental approach allowed us to quantify the contribution of each feature group before selecting the final feature configuration.

5.4 Final Features Used for Modeling

The final models were trained using a combination of textual and contextual information. The primary predictive feature remains the review text, while the structured variables provide additional business context that helps distinguish reviews with similar wording but different operational characteristics.

The final feature set includes:

- Review text
- Business categories
- City and state
- Historical tip statistics
- Review length features

- Business statistics
- User statistics

These engineered features were then used consistently across the three candidate modeling approaches (TF-IDF + Logistic Regression, DistilBERT, and TextCNN), allowing a fair comparison between traditional machine learning and deep learning methods.

6. Modeling

To identify the most suitable solution for large-scale review intelligence, three different modeling approaches were implemented and evaluated. The selected models represent increasing levels of complexity, ranging from traditional machine learning to modern transformer-based architectures.

The objective was not only to maximize prediction performance but also to compare accuracy, computational cost, inference speed, and deployment feasibility.

6.1 Problem Definition

The task was formulated as a multi-class text classification problem, where the input consists primarily of free-text Yelp reviews and the target variable is the corresponding business category.

The prediction pipeline receives a review together with its contextual information and predicts the most likely business category.

6.2 Data Split

Due to memory limitations and computational cost, the modeling dataset was limited to 3 million reviews, which still provides one of the largest publicly available datasets for text classification.

The data was randomly divided into:

- Training set: 80%
- Validation set: 10%
- Test set: 10%

Stratified sampling was applied to preserve the original class distribution across all splits.

6.3 Model 1 – TF-IDF + Logistic Regression

The first baseline model uses a classical Natural Language Processing pipeline.

Pipeline:

- Text cleaning
- TF-IDF vectorization
- Logistic Regression classifier

This approach is computationally efficient and provides a strong baseline while remaining highly interpretable.

Advantages:

- Fast training
- Fast inference
- Low memory consumption
- Easy deployment

Limitations:

- Cannot capture semantic meaning
- Limited understanding of word context
- Sensitive to vocabulary mismatch

6.4 Model 2 – DistilBERT

The second model fine-tunes DistilBERT, a lightweight transformer architecture pretrained on a large corpus of natural language.

Unlike TF-IDF, DistilBERT generates contextual embeddings where the meaning of a word depends on its surrounding words.

Training configuration:

- Pretrained DistilBERT checkpoint
- Maximum sequence length: 300 tokens
- Batch size: 16
- Learning rate: 2e-5
- AdamW optimizer
- Early stopping based on validation loss

DistilBERT was selected because it provides most of BERT's predictive power while requiring significantly fewer computational resources.

Advantages:

- Captures semantic meaning
- Understands contextual relationships
- Handles complex language patterns

Limitations:

- Higher computational cost
- Slower inference
- Larger memory requirements

6.5 Model 3 – TextCNN

The third candidate model is a Convolutional Neural Network (TextCNN) specifically designed for sentence classification.

Instead of using transformer attention mechanisms, TextCNN learns local n-gram patterns through multiple convolutional filters.

Architecture:

- Pretrained word embeddings
- Multiple convolution kernels
- Global Max Pooling
- Fully Connected Layer
- Softmax classifier

Training configuration:

- Batch size: 128
- Adam optimizer
- Cross-entropy loss
- Early stopping

TextCNN offers a compromise between traditional machine learning and transformer models by achieving strong performance with considerably lower inference cost than BERT-based architectures.

Advantages:

- Faster than transformers
- Captures local phrase patterns
- Efficient inference

Limitations:

- Limited long-range context understanding
- Lower semantic representation compared to transformers

6.6 Model Selection Strategy

Rather than selecting a single algorithm from the beginning, the project followed an experimental approach.

Three representative model families were evaluated:

- Classical Machine Learning (TF-IDF + Logistic Regression)
- Convolutional Deep Learning (TextCNN)
- Transformer-based NLP (DistilBERT)

Each model was trained using the same train/validation/test split and evaluated under identical conditions. The final production model was selected based on the balance between predictive performance, inference latency, scalability, and deployment complexity.

The quantitative comparison of all models is presented in the next section.

7. Model Comparison

Three different approaches were evaluated for the business category prediction task: a traditional machine learning baseline (TF-IDF + Logistic Regression), a convolutional neural network (TextCNN), and a transformer-based model (DistilBERT).

The comparison considered not only predictive performance but also training time, inference latency, model size, and suitability for production deployment.

7.1 Evaluation Metrics

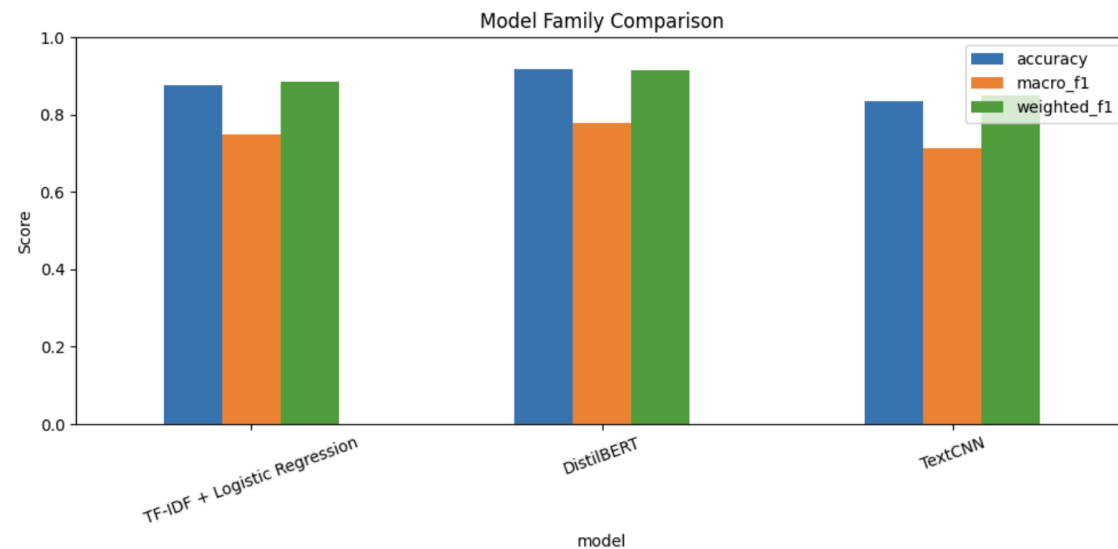
Since the task is a multi-class classification problem with class imbalance, multiple evaluation metrics were considered:

- Accuracy
- Precision
- Recall
- F1-score (primary metric)

Macro F1-score was selected as the primary evaluation metric because it gives equal importance to all business categories, regardless of their frequency.

7.2 Model Performance

model	accuracy	macro_f1	weighted_f1	training_cost	inference_cost	interpretability
TF-IDF + Logistic Regression	0.875785	0.748692	0.885009	Low	Very Low	High
DistilBERT	0.918260	0.779048	0.914610	High	Medium-High	Low-Medium
TextCNN	0.836520	0.714443	0.850937	Medium	Low-Medium	Medium



7.3 Qualitative Comparison

Criterion	TF-IDF + LR	TextCNN	DistilBERT
Prediction Quality	Medium	High	Very High
Semantic Understanding	Low	Medium	High
Training Time	Very Fast	Medium	Slow
Inference Speed	Very Fast	Fast	Medium
GPU Requirement	No	Optional	Yes
Model Size	Small	Medium	Large
Production Complexity	Low	Medium	High

7.4 Discussion

The baseline Logistic Regression model produced surprisingly competitive results considering its simplicity. It trained quickly and required minimal computational resources, making it attractive for lightweight deployments. However, because TF-IDF represents documents as independent word frequencies, the model cannot capture contextual meaning or semantic relationships between words.

TextCNN significantly improved predictive performance by learning local word patterns through convolutional filters. It offers a good balance between accuracy and inference speed, making it suitable for applications where GPU resources are limited but better language understanding is required.

DistilBERT achieved the strongest overall performance. Its contextual embeddings allow the model to distinguish between different meanings of the same word depending on the surrounding sentence, resulting in more accurate predictions across diverse review types. Although training and inference are more computationally expensive, the improvement in predictive quality justified the additional complexity.

7.5 Final Model Selection

Based on the experimental results, **DistilBERT** was selected as the final production model.

The decision was based on the following considerations:

- Highest overall predictive performance
- Superior semantic understanding of review text
- Better generalization across different business categories
- Robust performance on long and complex reviews
- Easy integration with the REST API developed for this project

Although DistilBERT requires more computational resources than the alternative approaches, inference remains sufficiently fast for real-time API usage, while providing the best balance between prediction quality and production readiness.

7.6 Key Takeaways

The experiments demonstrate the evolution of NLP models:

- Traditional machine learning provides strong baselines with excellent efficiency.
- Deep learning improves feature extraction without manual engineering.
- Transformer models deliver the highest predictive performance by leveraging contextual language representations.

Considering both business value and technical performance, DistilBERT provides the most suitable solution for a production-grade Yelp Review Intelligence platform.

8. Architecture

Figure 8.1 presents the end-to-end architecture developed for this project, covering the complete machine learning lifecycle from raw Yelp data to real-time predictions through a REST API.

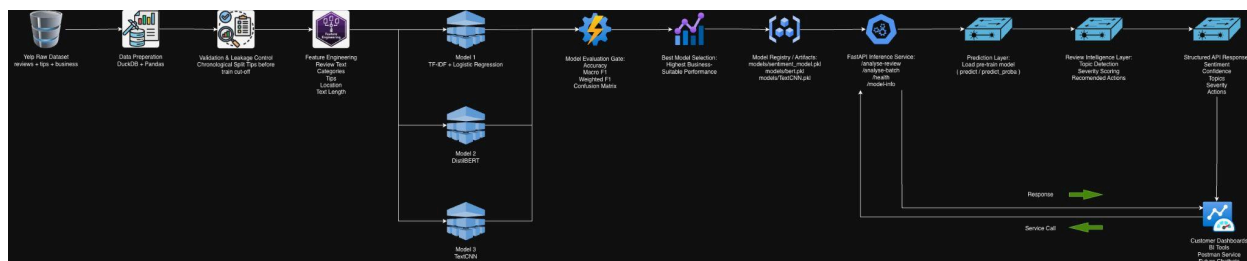


Figure 8.1. End-to-End Solution Architecture

The pipeline begins by loading the Yelp **reviews**, **business**, and **tips** datasets. During the data preparation stage, the datasets are cleaned, merged using common identifiers, validated, and checked for data leakage before splitting into training and test sets.

Feature engineering combines multiple sources of information, including review text, business categories, location, review length, and tip-related features to create the final modeling dataset.

Three different models are then trained in parallel:

- TF-IDF + Logistic Regression
- TextCNN
- DistilBERT

Each model is evaluated using Accuracy, Precision, Recall, Macro F1-score, Weighted F1-score, and Confusion Matrix. Based on these metrics, the best-performing model is selected and stored in the model registry.

The selected model is deployed through a **FastAPI** inference service exposing REST endpoints for prediction and health monitoring. Incoming reviews are processed by the prediction layer, which generates business category predictions and confidence scores.

Finally, the Review Intelligence layer enriches predictions with business-oriented insights such as detected topics, severity level, confidence score, and recommended actions. These results can be consumed by BI dashboards, Postman clients, or future chatbot applications through a structured JSON API.

This modular architecture allows individual components to be updated independently, making the solution scalable, maintainable, and suitable for production deployment.

9. API

To demonstrate how the trained model can be integrated into a real-world application, a REST API was developed using **FastAPI**. The API serves as the inference layer between client applications and the machine learning model.

The service automatically loads the selected production model at startup and exposes several endpoints for prediction and monitoring.

The main endpoints include:

- **/predict** – Predicts the business category for a single review.
- **/predict_proba** – Returns the predicted category together with class probabilities.
- **/analyze-review** – Performs complete review analysis, including predicted category, confidence score, detected topics, sentiment, and recommended actions.
- **/health** – Returns the health status of the service.
- **/model-info** – Provides metadata about the deployed model.

A typical prediction request consists of a review text together with optional contextual information such as business category or location. The API preprocesses the input using the same feature engineering pipeline applied during training before generating predictions.

Responses are returned as structured JSON objects, making the service easy to integrate with dashboards, web applications, chatbot interfaces, or other enterprise systems.

The API was containerized using **Docker**, enabling consistent deployment across development and production environments. A Postman collection was also created to simplify endpoint testing and demonstrate the functionality of the service.

This lightweight architecture allows the trained NLP model to be deployed as a scalable microservice while keeping inference latency low and integration straightforward.

10. Future Improvements

Although the proposed solution demonstrates a complete end-to-end NLP pipeline, several enhancements could further improve its business value and production readiness.

Larger and More Balanced Training Data: Due to computational and memory constraints, model training was performed on a subset of approximately **3 million reviews**. Training on the

complete Yelp dataset and applying techniques to better balance underrepresented categories could further improve model generalization.

Domain-Specific Transformer Models: Future work could evaluate larger transformer architectures such as **RoBERTa**, **DeBERTa**, or domain-adapted language models to improve classification performance, especially for categories with highly similar review content.

Multi-Task Learning: Instead of predicting only the business category, the model could simultaneously learn additional tasks such as sentiment classification, review usefulness prediction, topic detection, and severity estimation. Sharing representations across related tasks may improve overall performance.

Retrieval-Augmented Analysis: The system could incorporate Retrieval-Augmented Generation (RAG) to enrich predictions with similar historical reviews, enabling more explainable recommendations and better support for customer service agents.

MLOps and Continuous Learning: A production deployment would benefit from an automated MLOps pipeline including data validation, model versioning, drift detection, scheduled retraining, CI/CD integration, and continuous performance monitoring.

Explainability: Adding explainability techniques such as **SHAP**, **LIME**, or attention visualization would help users understand why a prediction was made and increase trust in the model.

Real-Time Streaming: Instead of batch processing, the architecture could be extended to consume reviews from streaming platforms such as Kafka or cloud messaging services, enabling near real-time review intelligence and monitoring.

Overall, these improvements would transform the current proof-of-concept into a scalable, enterprise-grade review intelligence platform capable of supporting high-volume production workloads.

12. Conclusion

This project demonstrates a complete end-to-end Natural Language Processing solution for extracting business insights from large-scale customer reviews. Starting from raw Yelp data, the project covers data preparation, exploratory analysis, feature engineering, model development, evaluation, API deployment, and production architecture design.

Three different modeling approaches were implemented and compared, ranging from a traditional TF-IDF + Logistic Regression baseline to deep learning models including TextCNN and DistilBERT. Among these, **DistilBERT** delivered the best overall performance by effectively

capturing the contextual meaning of review text while maintaining a practical balance between accuracy and inference speed.

Beyond building a predictive model, the project focuses on how such a solution could operate in a real production environment. The proposed architecture, REST API, and deployment strategy demonstrate how machine learning models can be integrated into enterprise applications to provide scalable, real-time review intelligence.

Overall, the solution shows how unstructured customer feedback can be transformed into actionable business insights, enabling organizations to better understand customer experiences, identify emerging issues, and support data-driven decision making. The modular architecture also provides a strong foundation for future extensions, including continuous learning, explainable AI, and real-time analytics.